

Sentiment Analysis of Financial News

Group R2

Astha Shrestha
524857

Technical University Berlin
Berlin Germany

Camila Luna Vasconez
464097

Technical University Berlin
Berlin Germany

ABSTRACT

This project focuses on the sentiment analysis of financial news through the Financial PhraseBank dataset. After analyzing the problems related to class imbalances, sentence lengths, and frequent usage of financial vocabulary, we employed a moderate-level pipeline for pre-processing that retains numbers and negation. Classic machine learning models utilizing Bag-of-Words and TF-IDF representations were compared to the fine-tuned DistilBERT model under multiclass and binary classification. Naive Bayes along with Bag-of-Words performed the best among classic models, whereas DistilBERT demonstrated the best performance overall. The challenge was to distinguish neutral news from positive or negative financial sentiment.

KEYWORDS

financial sentiment analysis, text classification, Naive Bayes, neural network, DistilBERT, textual similarity

1 Introduction and Data

This project focuses on sentiment analysis of financial news sentences. The goal is to classify each sentence as positive, neutral, or negative and to understand how different preprocessing methods, feature representations, and models affect the results. We used the Financial PhraseBank dataset, especially the Sentences_50Agree.txt file, where each line is written in the format `sentence@sentiment[1]`.

Financial sentiment classification is challenging because many sentences are short and written in a factual style. A sentence may contain financial words such as “profit”, “sales”, or “growth”, but these words do not always indicate the same sentiment in every context. Therefore, the project compares classical machine learning models using Bag-of-Words and TF-IDF features with a fine-tuned DistilBERT model [2], [3]. We also compare multiclass classification with a binary setting where the neutral class is removed.

In addition to classification, we analyze textual similarity between positive sentences by averaging word vectors and computing cosine similarity manually. Overall, the project focuses not only on final accuracy, but also on model behavior, error patterns, and the effect of methodological choices.

2 Exploratory Analysis and Preprocessing

As a part of the exploratory analysis, we analyzed the structure of the Sentences_50Agree.txt dataset. As mentioned earlier, this dataset consists of short sentences from financial news sources and includes three classes – positive, neutral, and negative sentiments. We studied the sentiment balance, lengths of sentences, vocabularies of each class, and frequent words/bigrams. Since financial news usually contain a lot of neutral factual sentences, there is a high risk of having a class imbalance problem. That is why, besides the accuracy, it is also necessary to evaluate the models by the macro F1 and class-wise precision/recall metrics.

We have found out that most of the texts in the dataset are quite short. It makes their sentiment classification problematic since the model does not receive much context. Most of the time, the sentiment of the whole sentence relies on just a few financial terms like “profit”, “loss”, “rise”, “fall”, or “sales”. Unfortunately, they cannot be the only indicator of the sentiment of a particular sentence as they may appear in any context. E.g., mentioning “profit” cannot determine whether the sentence has a positive or negative sentiment if it talks about the decline.

Regarding the text cleaning process, we applied a medium approach to the data. Lowercasing, tokenizing, and removing punctuation have been done, as well as replacing all numbers by one generic token since numbers are very popular in financial news and help avoid many rare features based on their exact values.

Stop-word removal has been applied with some caution. Regular words of no emotional or semantic importance like articles and prepositions were removed from the dataset. At the same time, words indicating the negation of sentiments like “not”, “no”, “never”, or “without” have been left because they can affect the sentiment interpretation of the whole sentence. Finally, we decided not to apply lemmatization and stemming for our task as they are rather aggressive processes that may eliminate some useful differences between similar domain-specific terms.

3 Classical Sentiment Classification

We performed classical sentiment classification using an 80/20 stratified train-test split. Stratification was used to maintain the

original class distribution in the training and test sets. We explored two feature representations, Bag-of-Words and TF-IDF, and two models, Multinomial Naive Bayes and feed-forward neural network with MLPClassifier. The classical vectorizers and classifiers were implemented with scikit-learn [5].

In the multiclass setting, Naive Bayes with Bag-of-Words achieved an accuracy of 0.7134 and macro F1 score of 0.6519. The naive bayes with tfidf had an accuracy of 0.6794 and a macro f1-score of 0.4351. The feed-forward neural network with Bag-of-Words achieved an accuracy of 0.7000 and macro F1 score of 0.6428. The neural network with TF-IDF achieved an accuracy of 0.6897 and macro F1 score of 0.6327. Best classical multiclass model was Naive Bayes with Bag-of-Words.

The worst model was Naive Bayes with TF-IDF. It was not really low in accuracy but it had a bad macro F1 score because it almost never predicted the negative class correctly. This is why accuracy is not enough by itself for this dataset. Macro F1, which is equally important to all classes, and which is poorly performed in minority classes, is more informative.

We looked at precision, recall, F1-score, support and confusion matrices for all the models in addition to their accuracy and the macro F1. For the multiclass model, which was Naive Bayes with Bag-of-Words the precision was 0.67 and the recall was 0.64. The confusion matrix for Bayes with Bag-of-Words is shown in Figure 1.

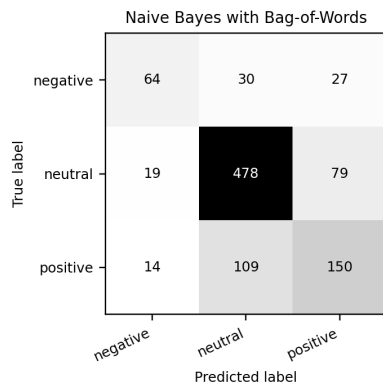


Figure 1 Confusion matrix of the best classical multiclass model, Naive Bayes with Bag-of-Words.

We performed error analysis by looking at the incorrect predictions made by the best classical model, Naive Bayes with Bag-of-Words. The model made 278 incorrect predictions on the test set. The most common error was positive sentences being predicted as neutral. This is because many positive financial sentences are written as factual announcements and do not contain highly emotional words. Another frequent mistake was the classification of neutral sentences as positive, especially those containing words like profit, sales, gains or figures of money. These errors show a limitation of word-count models that they do not fully understand if a financial

term is positive in context. The confusion matrix supports this error analysis. The largest off-diagonal value is positive predicted as neutral (109 cases), followed by neutral predicted as positive (79 cases).

We also re-trained the classification task as a binary classification task by removing all neutral examples. This yielded 1,967 examples, 1,363 positive, 604 negative. In the binary setting, Naive Bayes with Bag-of-Words achieved 0.8071 accuracy and a macro F1 score of 0.7734. Naive Bayes with TF-IDF yielded an accuracy of 0.7589 and a macro F1 score of 0.6069. The neural network with Bag-of-Words achieved an accuracy of 0.7995 and a macro F1 score of 0.7660, and the neural network with TF-IDF achieved an accuracy of 0.7995 and a macro F1 score of 0.7650. The neural network with Bag-of-Words had accuracy of 0.7995 and macro F1 score of 0.7660, and the neural network with TF-IDF had accuracy of 0.7995 and macro F1 score of 0.7650. The binary task was easier than the multiclass task because the model did not have to differentiate between neutral facts and positive/negative news. However, negative examples remained harder, especially for the TF-IDF Naive Bayes model, which had low negative recall.

4 Textual Similarity

As part of Textual Similarity, 15 sentences with positive sentiment have been extracted from the dataset. While the goal of the previous tasks has been classification, the purpose of this task was to analyze how similar positive financial news sentences are to each other. Sentences have been represented with sentence-level vectors as average word vectors that make up a sentence.

Sentence vectors have been compared using cosine similarity. The calculation of this metric has been done through manual implementation: cosine similarity has been determined by dividing dot product of two vectors by their vector product. This allowed us ensure full control over cosine similarity calculation process rather than relying on pre-defined functions.

Most of the time sentence pairs with high cosine similarity had overlapping financial semantics, i.e., sentences discussing sales improvement, growth of profit, market growth, signing contracts, etc. Their cosine similarity scores were high due to similarities in their averaged word vectors that share semantic information.

Least similar pairs were still positive in sentiment, yet referred to different aspects of a business. They included a combination such as sentence about agreements between businesses versus sentence about earnings or market growth. As shown by this example, similarity of sentiment does not necessarily equal similarity of semantics. Two sentences might both express positive sentiment, however, they do not have to be semantically related since they refer to unrelated events.

This approach of calculating sentence-level vectors via averaging word vectors has several limitations. First, word order plays an

important role in financial language. It also means that certain phrases may carry important financial meaning. At the same time, even individual words, such as “profit”, may appear in both positive and negative contexts. Thus, while averaged word vectors are easy to interpret and calculate, they cannot capture all of sentence semantics.

5 Pre-trained Language Model

For the pre-trained language model task, we fine-tuned distilbert-base-uncased, a smaller and faster BERT-style transformer model. We fine-tuned DistilBERT-base-uncased for the pre-trained language model task. DistilBERT is a smaller and faster BERT style transformer model [6]. In contrast to the classical models, DistilBERT does not rely on Bag-of-Words or TF-IDF features. Instead, the original sentence text is sent to the DistilBERT tokenizer, which produces subword tokens and attention masks. The model was trained with an 80/20 stratified split, one epoch, batch size 16, learning rate 2e-5 and maximum sequence length 64. The MPS device was used on the Mac if it was present.

For the multiclass setting, DistilBERT achieves an accuracy of 0.8103 and macro F1 score of 0.7954. This was significantly better than the top classical multiclass model, Naive Bayes with Bag of Words, which achieved an accuracy of 0.7134 and macro F1 score of 0.6519. With DistilBERT, the accuracy of 0.9518 and macro F1 score of 0.9442 were achieved in the binary setting. This also outperformed the best classical binary model, Naive Bayes with Bag-of-Words, which scored 0.8071 in accuracy and 0.7734 in macro F1.

The transformer model probably did better because it benefits from contextual representations. In the domain of financial sentiment analysis, words such as profit, growth, sales or loss cannot be interpreted reliably in isolation. They mean whatever is in the sentence around them. DistilBERT can represent words in different ways depending on the context. Classical word-count methods are more independent of words. However, the transformer model also has disadvantages. It requires more computation, additional packages, a downloaded pre-trained model, and longer training time. The exact scores of the transformer can also vary slightly from run to run due to the randomness in the training of the neural network, especially when using hardware acceleration.

The detailed classification report tells us more about how the model behaves. In the multiclass setting, the DistilBERT model got F1-scores of 0.78 for negative 0.85 for neutral and 0.75 for positive. In the binary setting, the DistilBERT model did better getting F1-scores of 0.92 for negative and 0.96 for positive. This shows that not having the neutral class made the task easier for the DistilBERT model. See figure 2.

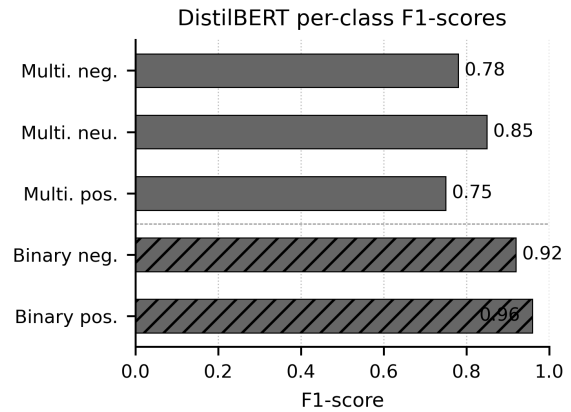


Figure 2 Per-class F1-scores of DistilBERT in the multiclass and binary sentiment classification settings.

As can be seen from the above results on the level of classes, the improvement of the transformer was mainly achieved for the minority negative class. Nevertheless, the positive class remains the class with the lowest multiclass F1-score, meaning that there might be cases where it was problematic to label a financial sentence if this sentence had been used as an actual announcement by a firm. The remaining typical error pattern is therefore likely still positive-neutral ambiguity, although DistilBERT reduces this problem compared with word-count models. There is also an additional computational disadvantage that does not apply only to training. These scores for each class show that the transformer model got better at handling sentences that were not as common.

6 Conclusion

In conclusion, the project proves that financial sentiment analysis is heavily dependent on class imbalance, short texts, and domain-specific language.

Classical models represent a helpful baseline with the two best-performing classical methods being Naïve Bayes and Bag-of-Words. Yet, the results obtained with DistilBERT model are better than those achieved with classical methods, which indicates that more sophisticated context-aware representations are more appropriate for capturing sentiments in finance-related texts.

REFERENCES

- [1] Pekka Malo, Ankur Sinha, Pyry Takala, Pekka Korhonen, and Jyrki Wallenius. 2013. Good debt or bad debt: Detecting semantic orientations in economic texts. *Journal of the Association for Information Science and Technology*.
- [2] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay. 2011. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* 12, 2825–2830.
- [3] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2019. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*.
- [4] Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest,

and Alexander M. Rush. 2020. Transformers: State-of-the-Art Natural Language Processing. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, 38–45.

- [5] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830
- [6] Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter. arXiv. <https://arxiv.org/abs/1910.01108>